

AI-Mediated Collaborative Cognition: Designing Systems That Preserve Reasoning in the Age of LLMs*

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Abstract

Large Language Models (LLMs) fundamentally weaken the relationship between student reasoning and observable output in computing education. When correct answers can be generated with minimal cognitive effort, traditional signals of understanding degrade, creating the risk of *cognitive bypass*—the production of correct results without engagement in the reasoning processes those results are intended to reflect. This paper argues that addressing this challenge requires a shift from tool-based integration of AI toward system-level design that enforces reasoning as a necessary condition for progress. Rather than treating AI as an open-ended assistant, we propose a constraint-based approach in which collaboration structure, interaction rules, and AI behavior are explicitly controlled to preserve cognitive engagement. We present coLearn-AI as an instance of this design approach. The system enforces role-based collaboration and turn-taking, constrains AI interaction to a scaffolding role, and maintains an append-only record of student interaction that

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captures intermediate reasoning steps. It supports natural-language responses, executable Python and C++ code, and analysis of program output, allowing AI feedback to operate across common computing artifacts while being constrained toward guidance rather than direct solution generation. Preliminary classroom observations suggest more consistently distributed participation, less divide-and-conquer behavior, and greater instructor visibility into student reasoning processes. This work contributes a design perspective in which preserving reasoning requires not more capable AI, but carefully engineered constraints on how AI and students interact.

1 Introduction

The rapid adoption of Large Language Models (LLMs) in computing education has created a fundamental tension: while these systems can enhance productivity, they also decouple observable output from the cognitive effort traditionally required to produce it.

This shift undermines both assessment and learning. If output is no longer a reliable proxy for reasoning, educators must reconsider how learning environments are structured. The central question becomes:

How can we design systems where AI strengthens, rather than bypasses, the learning process?

Learning theory suggests that reasoning cannot be inferred solely from final outcomes, but must be observed through the processes that produce them. In environments where AI can generate correct outputs independently, preserving these processes becomes a central design challenge.

Prior work, including our own, has focused on building systems that integrate AI into guided inquiry workflows. In this paper, we shift focus from system features to the underlying design problem introduced by LLMs: the breakdown of the relationship between reasoning and observable output. We argue that addressing this requires not additional AI capability, but system-level constraints that preserve reasoning as a necessary condition for progress.

This paper proposes a design-oriented solution grounded in learning theory and illustrated through a collaborative platform, coLearn-AI. Rather than introducing new AI capabilities, the system demonstrates how structured interaction constraints can preserve reasoning as a necessary condition for progress in AI-mediated learning environments.

2 Related Work

2.1 AI in Education

Artificial Intelligence (AI) has been widely explored as a tool to support learning through intelligent tutoring systems, adaptive feedback, and automated assessment. Recent work on Large Language Models (LLMs) emphasizes personalized instruction and immediate feedback [3, 5].

However, much of this work focuses on *individual learning*, where AI functions as a tutor or answer generator. While effective for efficiency and access, this approach risks reducing opportunities for productive struggle and deep cognitive engagement.

2.2 AI and Collective Intelligence

Recent work has begun to examine how AI can support *collective intelligence* in educational settings. Casebourne et al. [1] argue that AI can facilitate group reasoning, shared knowledge construction, and collaborative problem solving.

This perspective aligns with social constructivist views of learning as an interaction among individuals, tools, and representations, positioning AI as a participant in a broader cognitive system rather than a replacement for human reasoning.

2.3 Positioning This Work

Prior work on AI-supported collaboration emphasizes enabling interaction, often assuming that effective collaboration will emerge when tools are provided.

In contrast, we argue that collaboration must be *structurally enforced*. Rather than treating AI as an open-ended assistant, we impose explicit constraints on interaction, including turn-taking, role-based participation, constrained AI feedback, and persistent capture of intermediate work.

This represents a shift from *AI-supported collaboration* to *AI-constrained collaborative cognition*, where the system actively preserves the conditions required for reasoning.

3 Background and Learning Theory

This work is grounded in constructivist and social learning theory, which emphasize learning through active engagement and social interaction [6, 7].

This perspective aligns with social constructivist and distributed-cognition views of learning as an interaction among individuals, tools, and representa-

tions [7, 4], positioning AI as a participant in a broader cognitive system rather than a replacement for human reasoning.

The system’s collaborative structure is informed by Process-Oriented Guided Inquiry Learning (POGIL), which organizes students around structured inquiry and defined collaborative responsibilities [2].

4 The Problem: LLMs and Cognitive Bypass

Traditional models of learning implicitly rely on a stable relationship between cognitive effort, reasoning processes, and observable output. In these models, students engage in cognitive effort, which produces reasoning, and this reasoning in turn generates observable artifacts such as written answers or code. Educators then evaluate these outputs as proxies for understanding.

The emergence of Large Language Models (LLMs) fundamentally disrupts this relationship. LLMs enable the direct generation of correct or plausible outputs without requiring the learner to engage in the underlying reasoning processes. As a result, the observable artifact—previously a proxy for cognition—can no longer be assumed to reflect the student’s understanding.

This decoupling has significant implications for both learning and assessment. First, assessment signals degrade: correct answers no longer reliably indicate comprehension. Second, students may bypass productive struggle, a key mechanism through which cognitive restructuring occurs in constructivist learning theory. Third, opportunities for deep learning diminish, as the cognitive processes required for durable understanding are no longer consistently engaged.

We define this phenomenon as *cognitive bypass*: the ability to produce correct outputs without engaging in the reasoning processes those outputs are meant to reflect.

This challenge is not simply a matter of restricting AI usage, but of redesigning learning environments so that reasoning remains necessary for progress. Rather than treating AI as an external tool, we must embed constraints within the system itself that preserve the cognitive conditions required for learning. This shift is illustrated in Figure 1.

5 Design Principles

To address cognitive bypass, learning environments must enforce conditions in which reasoning is required for progress.

First, systems must require engagement in intermediate reasoning. Constructivist theory emphasizes the role of productive struggle and cognitive conflict in learning; therefore, systems should prevent direct answer generation

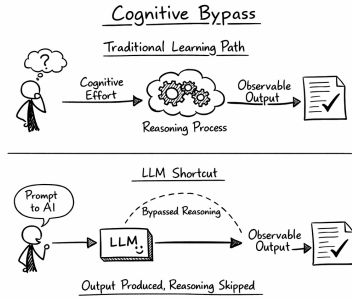


Figure 1: Cognitive bypass: LLMs weaken the connection between reasoning and observable output. In the traditional model (top), cognitive effort produces reasoning that leads to output. In the LLM-mediated case (bottom), output may be produced without engaging in the underlying reasoning process.

and instead require students to articulate partial explanations, predictions, or justifications before receiving feedback.

Second, collaboration must be structurally enforced. Social constructivist perspectives highlight that reasoning develops through interaction, yet collaboration is often uneven in practice. Mechanisms such as role assignment and turn-taking ensure that participation is distributed and visible.

Third, systems must preserve process. Because reasoning unfolds over time, capturing only final answers obscures how understanding develops. Systems should record intermediate responses, revisions, and feedback to support analysis of student thinking.

Fourth, AI must be constrained to a scaffolding role. While LLMs can provide useful guidance, unconstrained interaction allows them to replace rather than support reasoning. AI should instead prompt, question, and guide student reasoning without providing complete solutions.

Finally, systems must enforce participation. In group settings, some students may disengage or defer to others; requiring active contribution ensures that all participants engage in the reasoning process.

Together, these principles define a shift from permissive AI integration toward environments that actively preserve the cognitive conditions required for learning.

6 Persistent Epistemic Trace

To support reasoning as a first-class outcome, systems must capture not only final answers, but how those results are developed. We define a *persistent epistemic trace* as an append-only record of student responses, revisions, AI feedback, and executable artifacts.

Each submission produces a new entry rather than replacing prior work, preserving the sequence of reasoning over time. This enables instructors to examine how understanding develops and how participation is distributed within groups.

The trace provides instructors with process evidence for examining how answers developed and identifying possible cognitive bypass, while also supporting analysis of reasoning trajectories across attempts.

7 System Design: coLearn-AI

coLearn-AI is a web-based platform that operationalizes the design principles described above through a set of enforced interaction constraints. Rather than providing an open-ended interface to AI, the system structures how students interact with both each other and the AI, ensuring that reasoning remains central to the learning process.

Each student is assigned a fixed role for the duration of an activity, defining that student’s responsibilities within the group. Submission control, however, rotates among group members: at any given time, only one student may submit the group’s response. This combination preserves stable collaborative responsibilities while requiring every student to participate directly.

AI interaction is deliberately limited to a scaffolding role. Instead of generating solutions, the system uses AI to provide prompts, feedback, and probing questions that guide student reasoning without replacing it. The system supports multiple forms of student expression, including natural language responses, executable program code (e.g., Python and C++), and program output. AI-mediated feedback operates across these modalities, allowing the system to evaluate and guide not only written explanations, but also computational artifacts and their behavior. This multi-modal interaction is incorporated into the epistemic trace, enabling the system to capture reasoning as it unfolds across textual, procedural, and executable representations.

All interactions are persistently recorded as part of the system’s epistemic trace, capturing not only final answers but also intermediate responses, revisions, and AI interventions. This ensures that the process of reasoning remains visible and analyzable.

Together, these constraints transform collaboration from an optional activity into a structured process enforced by the system itself.

7.1 Example Interaction

Consider an activity in which students examine a Python loop, predict its output, and explain how the loop variables produce that result. The activity is displayed to every member of the group, but only the student currently holding submission control can enter and submit the group’s response. This structure requires the students to discuss and agree on an explanation before it is submitted.

Suppose the group correctly predicts the output but does not adequately explain how the loop variable changes. Rather than supplying the missing explanation, coLearn-AI might respond, “How does the value of `i` change during each iteration, and how does that change affect the value printed?” The group then discusses the prompt and submits a revised answer. This cycle continues until the response satisfies the instructor-defined criteria or the system determines that instructor intervention may be needed.

The original response, AI feedback, and subsequent revisions are retained in the interaction record, allowing instructors to examine how the group’s reasoning developed. After the question is completed, submission control rotates to another group member.

A publicly accessible demonstration of coLearn-AI is available at <https://colearn-ai.com/demo/aied2026>.

8 Operationalizing POGIL as a Protocol

Process-Oriented Guided Inquiry Learning (POGIL) [2] provides one example of a structured, collaborative learning approach that can be expressed in terms of system-enforced interaction constraints. Rather than relying on instructor facilitation alone, these constraints can be encoded directly within the platform.

In this view, roles define stable responsibilities, structured activities guide inquiry, submission control rotates among participants, and participation is enforced through the system. While POGIL provides the immediate pedagogical model, the same approach generalizes to other models that require coordinated reasoning and controlled interaction.

9 Preliminary Observations

The system has been deployed extensively in three sections of an introductory programming course, with additional intermittent use in Data Structures, Ap-

plied Algorithms, and Software Development. These observations are informal and not derived from a controlled study, but they provide initial insight into how structured, AI-mediated collaboration affects student behavior.

Prior to this system, POGIL-style activities were conducted using shared Google Docs. In that setting, collaboration was often uneven, with one student frequently completing most of the work. A common pattern was “divide and conquer,” where students split tasks rather than engaging in shared reasoning. Additionally, instructors had limited visibility into individual contributions and the overall group process.

Instructor observations suggested more consistently distributed participation with coLearn-AI. Turn-taking required students to alternate responsibility, while the interaction history gave instructors greater visibility into how answers were developed. Students generally reported a positive experience with the system.

At the same time, new challenges emerged. Some students progressed more slowly and could delay group advancement, while others attempted to move ahead without full group alignment. These behaviors highlight the need for improved real-time awareness of group pacing and more refined mechanisms for balancing participation and progress.

These observations suggest that system-level constraints can influence collaborative behavior, but they also expose a design tension. AI guidance that is too restrictive may block progress, while overly permissive guidance may allow cognitive bypass. Similarly, fixed roles and rotating submission control distribute responsibility but can allow a slower participant to delay the group. Balancing reasoning, participation, and progress remains an open problem.

10 Future Work

Several directions for future work emerge from this system and its deployment.

First, controlled empirical studies are needed to evaluate the impact of structured, AI-mediated collaboration on learning outcomes. In particular, the epistemic trace provides a rich source of data for analyzing reasoning processes over time and comparing them across different instructional conditions.

Second, improving the design and management of AI guidance is a critical area for further development. This includes externalizing prompt definitions from system code, enabling reuse of course- and topic-level guidance, and developing tools that allow instructors to iteratively test and refine prompts more efficiently.

Finally, adaptive mechanisms for group formation, pacing, and participation balancing may help address observed challenges in coordinating collaborative work across students with different working speeds and engagement patterns.

11 Conclusion

Large Language Models fundamentally alter the relationship between cognitive effort, reasoning, and observable output, weakening traditional assumptions about how learning can be assessed and supported. This shift introduces the risk of cognitive bypass, where correct answers can be produced without engaging in the reasoning processes that underpin understanding.

In response, this work argues that preserving learning in AI-mediated environments requires system-level intervention. Rather than treating AI as an external tool, we embed constraints directly into the learning environment to ensure that reasoning remains necessary for progress. The coLearn-AI platform demonstrates how structured interaction, constrained AI guidance, and persistent epistemic traces can work together to support collaborative reasoning.

The central contribution of this work is a design approach that integrates learning theory, system constraints, and AI into a unified approach for computing education. While POGIL serves as one illustrative example, the approach generalizes to other structured pedagogical models that depend on coordinated reasoning and controlled interaction.

As AI continues to reshape educational practice, the challenge is not simply how to use these systems, but how to design environments in which they strengthen, rather than replace, the cognitive processes essential for learning.

References

- [1] Imogen Casebourne et al. “Using AI to Support Education for Collective Intelligence”. In: *International Journal of Artificial Intelligence in Education* 35.3 (2025), pp. 1597–1629. DOI: 10.1007/s40593-024-00437-7.
- [2] David M. Hanson. *Instructor’s Guide to Process-Oriented Guided-Inquiry Learning*. Pacific Crest, 2013.
- [3] Wayne Holmes, Maya Bialik, and Charles Fadel. *Artificial Intelligence in Education: Promises and Implications for Teaching and Learning*. Center for Curriculum Redesign, 2019.
- [4] Edwin Hutchins. *Cognition in the Wild*. MIT Press, 1995.
- [5] OpenAI. “GPT-4 Technical Report”. In: *arXiv abs/2303.08774* (2023). DOI: 10.48550/arXiv.2303.08774. arXiv: 2303.08774 [cs.CL].
- [6] Jean Piaget. *The Origins of Intelligence in Children*. International Universities Press, 1952.
- [7] Lev S. Vygotsky. *Mind in Society: The Development of Higher Psychological Processes*. Harvard University Press, 1978.